Durham University

Computational Modelling in the Humanities and Social Sciences

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Summative Assignment 2026

Due: 12 January 2027

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The Four Rules of CompMod (Computational Modelling)

1. You do not plagiarize in CompMod
2. *You do not plagiarize in CompMod*
3. You must follow the assignment specification
4. Extensions and mitigations are handled by the CS undergraduate office

Not following the rules of CompMod means you risk failing CompMod

- Last 5 years: *all* failing submissions failed to follow one or more of rules 1-4
- Not following the rules means *you risk failing the module*

Specification

In this assignment, you will develop and critically evaluate one or more computational models appropriate to a specific humanities or social sciences modelling task of your own choosing. The task must relate directly to at least one of the technical topics covered in this module.

You will submit a written report of up to 3000 words describing your chosen task, sources of data, implementation, and evaluation together with documented code used to carry out the task. Your report must:

1. Begin with a title and introduction where you clearly state which model(s)/method(s) are used. No abstract is required.
2. Clearly state and motivate your chosen problem and research question(s). Provide sufficient background information to contextualize this problem for the reader.
3. Document sources of data, and briefly explain any relevant modules or toolkits that made significant contributions to the implementation/analysis (e.g. CoreNLP, Stanza, Gensim, QGIS, etc.).
4. Describe the chosen model(s) and implementation.
5. Summarize your results.
6. Critically evaluate the model(s). Depending on the task, this will typically involve both formal metrics of success (e.g. accuracy, precision, recall, etc. as appropriate) and critical analysis of the adequacy of the modelling (e.g. what are the assumptions the models rely on, and do they all hold; are there factors or biases that might invalidate conclusions drawn, etc.), as well as comparison with external data and/or published research on the topic.
7. Based on the evaluation, state and discuss what conclusions can be drawn from the model(s) with respect to the stated research question(s).

Assignment specification

Example task

Based upon data from Wikidata and using Geographic Information Systems, investigate the geographic distribution of places of birth of UK members of parliament over time, from the first Parliament of the United Kingdom (1801-1802) to the current parliament. Use data obtained from Wikidata combined with data from other sources to:

- Investigate regional biases: are there particular places or regions in which members of parliament are more likely to have been born?
- Do these biases change in any interpretably meaningful ways over time?
- What factors may explain the observed results? Can specific hypotheses about these be tested?
- How adequate is the data collected from Wikidata for answering these questions?

For further inspiration, please refer to the academic papers cited and discussed during lectures / provided as recommended readings and linked to on Ultra. You are encouraged to discuss your proposed task with the lecturer either by e-mail or during office hours.

Assignment specification

In this assignment, please note that:

- You are encouraged to reuse existing code – this includes open-source modules and toolkits, Stack Overflow responses, code described in online tutorials, etc. When doing so, you must clearly indicate what you have reused and exactly where it came from (including a URL for an online source). You should document smaller instances of reuse (e.g. short code examples from Stack Overflow) as comments in your code, and longer ones both in your code and as citations to the source in your report. Marks will be awarded for the original parts of your work (i.e. your extensions to and adaptations or developments of any reused material).
- Marks will not be awarded under the "Technical depth/clarity of implementation" sections for parts of your code that reimplement code available in widely used libraries or modules.

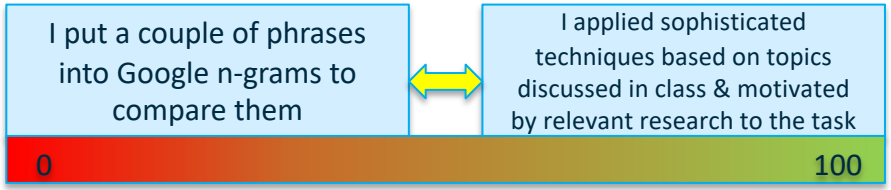
Assignment specification

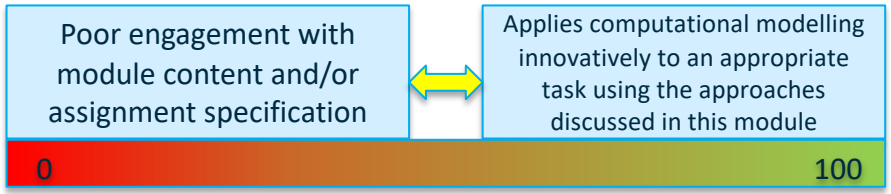
- You may use any programming language suitable for the task; use of Python 3 is recommended where possible.
- In writing your report, you should assume that the reader is familiar with everything covered during the lectures for this module. Concepts and techniques important to your implementation which were *not* introduced during the module should be explained.
- The word limit is exclusive of tables, figures, and references. You may include a maximum of 20 figures. Please use a 12pt font, single-spacing, and A4 pages.

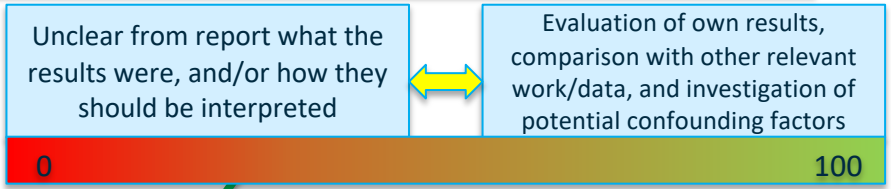
Submission (2 files)

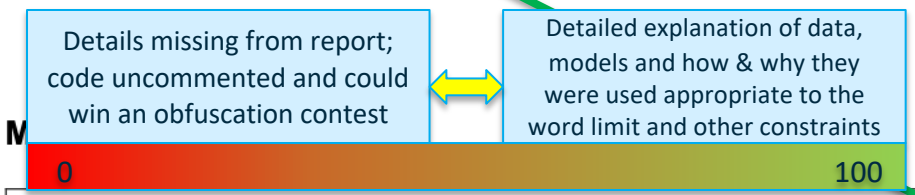
- A written report as a PDF file.
- An archive (.zip / .tar.gz) containing the code for your implementation, including instructions for how to run it in a README.TXT file. If any data that your code relies on exceeds 10Mb in size and is publicly available, include instructions in this file for how to obtain it. Your code is not marked separately, but serves as supporting evidence for criteria of the marking scheme (below): provided your code is suitably documented, it will be taken into account in assessing the technical depth and clarity of your implementation.

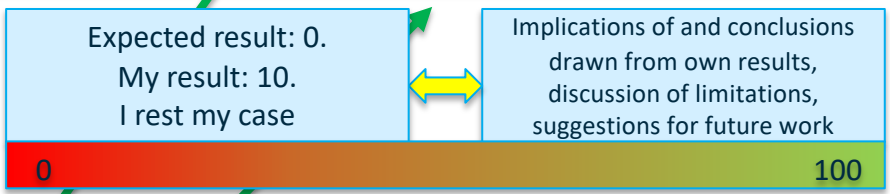
Marking scheme





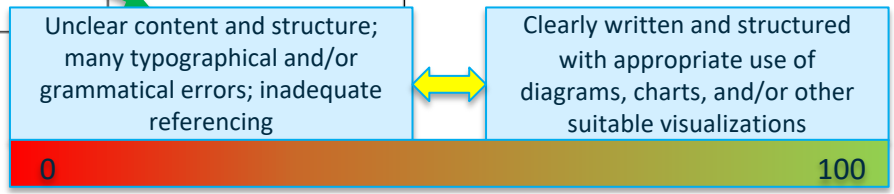






M

	Marks	Assessed by
1. Originality and appropriateness of topic	10	Report and code
2. Technical depth of implementation	25	Report and code
3. Clarity of implementation	20	Report and code
4. Evaluation	20	Report and code
5. Conclusions	20	Report
6. Presentation (appropriate academic writing, structure, and referencing)	5	Report



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Reminder: 6.2.4: Academic Misconduct

Academic Misconduct is a major offence which will be dealt with in accordance with the University's General Regulation IV – Discipline, and which may lead to expulsion from the University.

1. Academic Misconduct will be penalised, and takes various forms including:

a. plagiarism: the copying or paraphrasing of other people's work or ideas without full acknowledgment, which amounts to the presentation of someone else's work or ideas as one's own. Plagiarism includes the following:

(i) Collusion, which is when students work together in an unauthorised way to develop a submission for an assessment where such input is not permitted. Collusion does not depend on a fee being paid for the work;

(ii) Contract Cheating, which is when when a student submits work for assessment where they have used one or more of a range of services provided by a third party. The third party contract with the student can include payment or other favours, but this is not always the case. 'Services' may include essays or other types of assignments, conducting research, impersonation in exams and other forms of unfair assistance for completing assessed work. 'Third parties' include web-based companies or auction sites (essay mills), sharing websites (including essay banks), or an individual such as a lecturer, colleague, friend or relative. This applies in any case where the third party makes a contribution to the work of the student, such that there is reasonable doubt as to whose work the assessment represents;

(iii) Multiple submission, which is the inappropriate submission on multiple occasion of the same or substantially the same work of one's own for summative assessment in connection with an academic award;

b. **Impersonation:** presenting work on behalf of someone else as if it were the work of the other individual (this can be linked to collusion, or to cheating in examinations, and may include the facilitation of plagiarism through publication);

c. **Cheating:** using any inappropriate or unauthorised means to achieve credit for a piece of coursework or an examination answer. Cheating includes the following

d. **Use of inadmissible material**: using any material which is not permitted to achieve credit for a piece of coursework or an examination answer (which may also include falsification of results or evidence, or use of unethical research methods).

Doing any of these things is an easy way to get a 0 for this module!

If unsure whether something is acceptable, ask first!

Learning and Teaching Handbook

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Plagiarism and how to avoid it

"Copying or paraphrasing of other people's work or ideas *without full acknowledgment*"

- If you copy *anything* from *anywhere*, cite the source *where this is done*
  - This includes your report, *and* your code
    - Do not just give a list of references without stating what was copied
  - Code citations must include a URL *and a description of what was copied*
    - Do not just give a list of URLs without stating clearly what was copied
- Remember: marks are awarded for good referencing!
  - You cannot lose marks by having too many citations
  - More about plagiarism: "Cite Them Right: The Essential Referencing Guide"

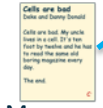
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Use of generative AI

"Copying or paraphrasing of other people's work or ideas *without full acknowledgment*" – including when distilled through an AI model!

- If you copy *anything* from *anywhere*, cite the source *where this is done*
  - This includes your report, *and* your code
  - If you use generative AI to help with coding, document this in your report *and* your code
  - If you use generative AI to help with writing, document this in your report
- In addition to describing this use in your report, you may also submit logs of prompts and responses in your code submission to fully document this
- More details: Computer Science Generative AI policy

Marking scheme:
Hitchhiker's summary guidance flowchart



My paper



My code

Did you submit on time?

No

Yes

Did you do the work yourself?

No

Yes

Acknowledged your sources?

No

Yes

Did you follow assignment spec?

No

Yes

Score going by marking scheme?

No deal

Game over man, game over!*

Take off and nuke submission from orbit**

This would be *bad*. Those fuzzy on the whole good-bad thing should refer to the progression flowcharts in the Faculty Handbook for full details

Your academic career may move forward into broad, sunlit uplands

Perfect submission with no identifiable issues of any kind

Perfect mark

Excellent

Pretty good

Kind of OK

Pass module

Good mark

Excellent mark

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* This means your submission is likely to receive a failing grade.

** It's the only way to be sure.

Main topics we will cover this term

So far:

- Classic models of text: N-grams, TF-IDF, cosine similarity
- (Today) Word embeddings

Still to come:

- Authorship attribution
- Topic modeling
- Text reuse
- Social Network Analysis
- Geographic Information Systems
- Agent-based modeling

Formative assignment

Briefly describe (max 250 words) your proposed modeling task

- See the coursework specification for an example task description
  - You **can** base your task on that one if you wish
- Submit via Ultra, due date: **19 November 2026**

Key questions you should consider in selecting your task:

- Does it connect to a specific real-world humanities/social science issue?
- Does it make use of a technical approach used in this module?
- What data is available that I can base my model on?

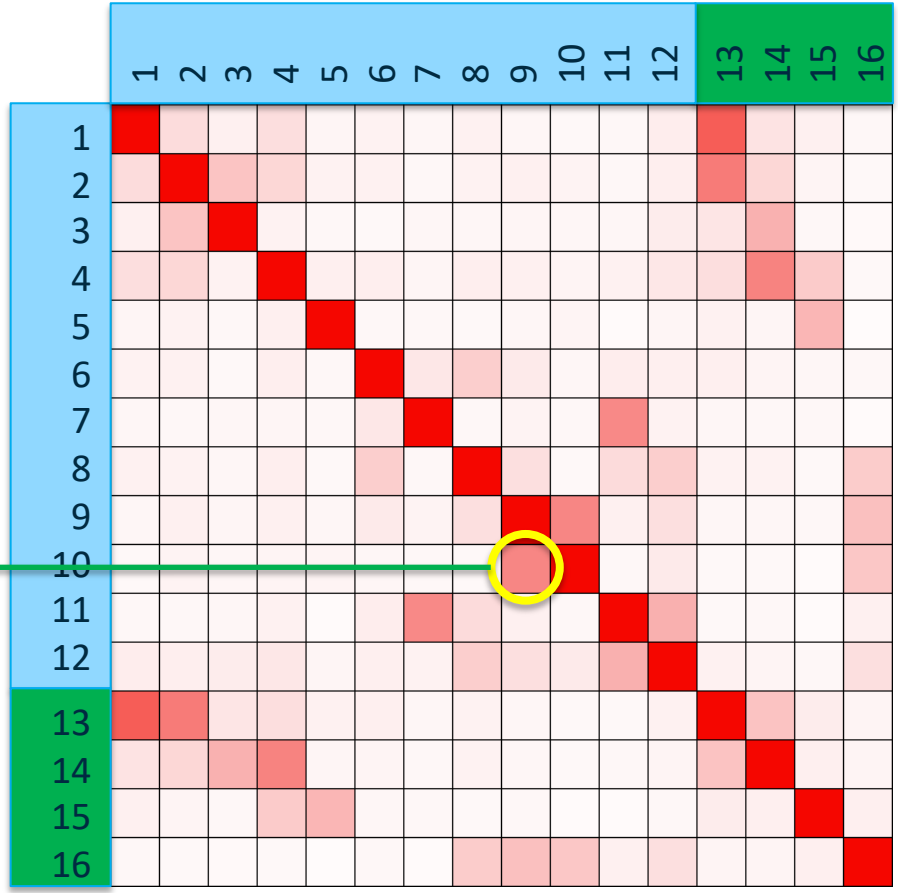
Word embeddings

Last time: Cosine similarity

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}}$$

Vocabulary sorted by A_iB_i

t_i	A_i	B_i	A_iB_i
Turtle	0.381	0.47	0.179
Mock	0.368	0.435	0.16
Gryphon	0.342	0.292	0.0999
Soup	0.175	0.042	0.0074
sea	0.0548	0.0311	0.0017
course	0.0328	0.0435	0.0014
school	0.0159	0.084	0.0013
Tis	0.0394	0.0261	0.001
replied	0.0312	0.0236	0.0007
Thank	0.0317	0.021	0.0007
old	0.0131	0.0435	0.0006
different	0.0372	0.0123	0.0005



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Semantic + syntactic similarity of terms

$T_1 \neq T_2$ => No similarity? For example:

a) "persuasive example of logical thinking about diagrams"

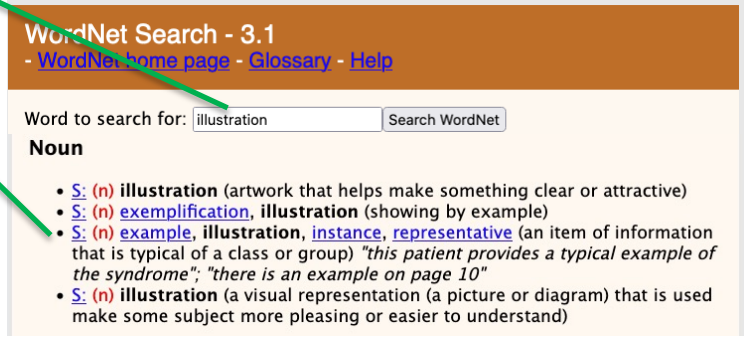
b) "compelling illustration relating to rational cognition regarding visualizations"

c) "some random discussion on blue cheese"

One of these things is not like the others 😉

$sim_{cosine}(a,b) = sim_{cosine}(a,c) = sim_{cosine}(b,c) = 0$

One approach: calculate similarity between tokens using a thesaurus (e.g. WordNet)



Semantic + syntactic similarity of terms

$T_1 \neq T_2$ => No similarity? For example:

a) "persuasive example of logical thinking about diagrams"

b) "compelling illustration relating to rational cognition regarding visualizations"

c) "some random discussion on blue cheese"

- S: (adj) **logical** (capable of or reflecting the capability for correct and valid reasoning) *"a logical mind"*
  - *see also*
    - S: (adj) analytic, analytical (of a proposition that is necessarily true independent of fact or experience) *"'all spinsters are unmarried' is an analytic proposition"*
    - S: (adj) coherent, consistent, **logical**, ordered (marked by an orderly, logical, and aesthetically consistent relation of parts) *"a coherent argument"*
    - S: (adj) synthetic, synthetical (of a proposition whose truth value is determined by observation or facts) *"'all men are arrogant' is a synthetic proposition"*
    - S: (adj) rational (consistent with or based on or using reason) *"rational behavior"; "a process of rational inference"; "rational thought"*
    - S: (adj) reasonable, sensible (showing reason or sound judgment) *"a sensible choice"; "a sensible person"*

WordNet Search - 3.1
- WordNet home page - Glossary - Help

Word to search for: illustration Search WordNet

Noun

- S: (n) **illustration** (artwork that helps make something clear or attractive)
- S: (n) exemplification, **illustration** (showing by example)
- S: (n) example, **illustration**, instance, representative (an item of information that is typical of a class or group) *"this patient provides a typical example of the syndrome"; "there is an example on page 10"*
- S: (n) **illustration** (a visual representation (a picture or diagram) that is used make some subject more pleasing or easier to understand)

Semantic + syntactic similarity of terms

$T_1 \neq T_2$ => No similarity?

"One-hot" encoding of terms

Problem: non-identical similarities ignored

this	example	is	another	example
1	0	0	0	0
0	1	0	0	1
0	0	1	0	0
0	0	0	1	0

Better representations:

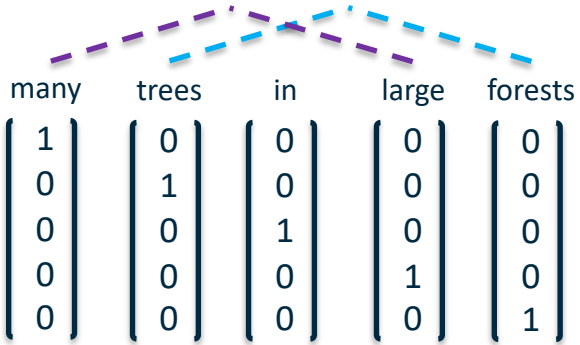
- Embeddings, word2vec, BERT, …

	D_1	D_2	D_3	V = [
	1	1	0	the
Semantic	1	1	0	power
	1	1	0	of
	1	1	1	example
	1	1	0	:
Syntactic	1	1	0	anthropological
	1	1	0	explorations
	1	1	0	in
Syntactic	1	1	0	persuasion
	1	2	2	,
Syntactic	1	1	0	evocation
	1	1	0	.
	0	1	1	and
Syntactic	0	1	0	imitation
Semantic+?syntactic	0	0	2	an
Syntactic	0	0	1	artificial
	0	0	1	about
?Semantic+syntactic	0	0	1	ant
Semantic+?syntactic	0	0	2	a
?Semantic+syntactic	0	0	1	rat
Semantic	0	0	1	powerful
?Semantic+syntactic	0	0	1	person]

Learning patterns from text

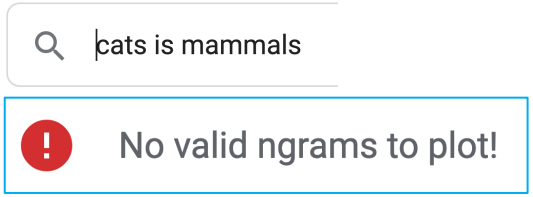
Infer from examples patterns in language content/form (i.e. syntax, semantics)

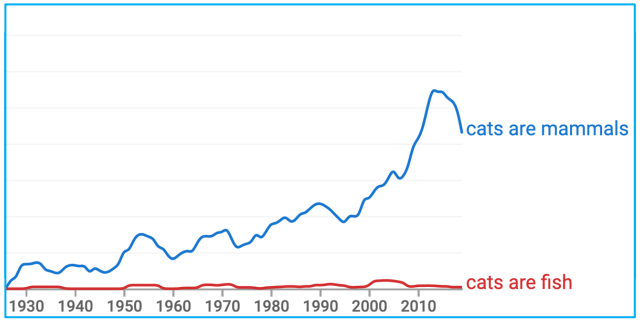
- Language is *infinitely flexible*
  - Compositionality: use of grammatical “rules”…
    - But also complex *semantic* rules, commonsense reasoning, …
- 1-hot encoding => “starting from scratch” – all words are equally different
  - Better approach: extract & use knowledge about “how language works”



similarity(trees, forests) = 0
similarity(many, large) = 0
…

Language models

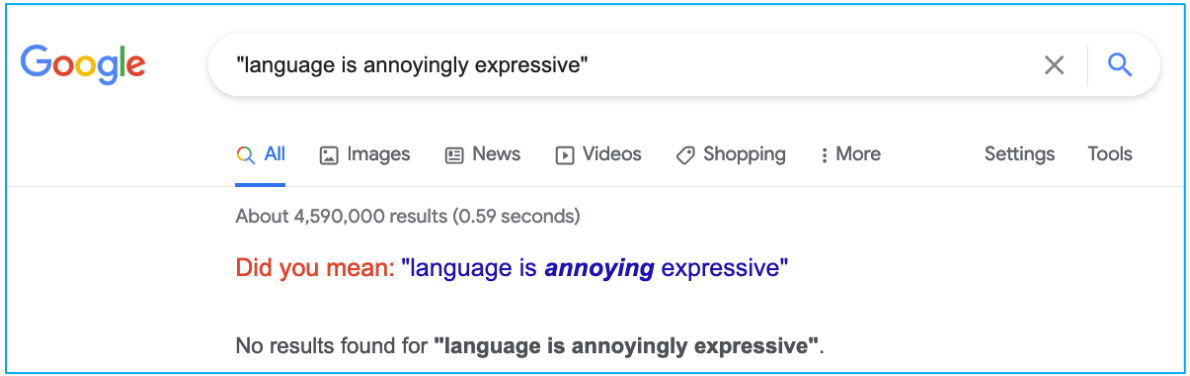




Probability distributions over sequences of words

- Which sequences of words are more/less valid/plausible
  - *Similar* to n-gram models in e.g. Google Books n-grams
  - But: should also predict validity of previously unattested sequences

Huge corpora available to train models



Better vector representations: motivation

Idea: predicting missing words relies on understanding of language

- I.e. understanding of complex mixture of semantics, syntax, context, etc.

Use this to create better vectorizations using easily generatable data

- Trivial to generate train+test data from unstructured text

- Kick the ball _____ the opponent's goal.
  1. is 2. into 3. of 4. between

- Apples grow on _____.
  1. time 2. average 3. trees 4. rocks

Ability to follow grammatical "rules" required to get the correct answer

Some syntactically correct answers are excluded by semantics & background knowledge

Better vector representations: motivation

…even greater proportion of the *information* does not yet exist in any…
… regular expressions with basic *information* from the annotation, …
…enabling the aggregation of *information* about entities independently from their…
…For example, broader *information* about the network structure of…

… works are a key source of *knowledge* about many aspects of…
…this process of annotation and *knowledge* extraction leads to entity records…
… often relies upon extracting *knowledge* from primary source materials…

…point in the annotation workflow, the *user* can also add claims about…
…the system uses prior *user* annotation decisions to offer...
…component is a client-side *user* interface for performing…
…this data to visually indicate to the *user* the location of probable entity…

Word-word co-occurrence matrix

Word \ Context	I	like	playing	football	enjoy	sports	Do
I	0	1	0	0	2	0	1
like	1	0	1	0	0	0	0
playing	0	1	0	1	0	0	0
football	0	0	1	0	1	0	0
enjoy	2	0	0	1	0	1	0
sports	0	0	0	0	1	0	0
Do	1	0	0	0	0	0	0

	Word vector
I	(0, 1, 0, 0, 2, 0, 1)
like	(1, 0, 1, 0, 0, 0, 0)
Playing	(0, 1, 0, 1, 0, 0, 0)
football	(0, 0, 1, 0, 1, 0, 0)
enjoy	(2, 0, 0, 1, 0, 1, 0)
sports	(0, 0, 0, 0, 1, 0, 0)
Do	(1, 0, 0, 0, 0, 0, 0)

- Input text (corpus): I like playing football. I enjoy sports. Do I enjoy football?
- Define the context of a word as 1 previous and 1 following word in a sentence
  - Many possible variations (e.g. TF-IDF instead of TF)

Sparse and dense word vectors

Using word-word co-occurrences will give vectors that are:

- High-dimensional – why?
- Sparse – why?

There are many ways we could reduce the dimensionality of these vectors

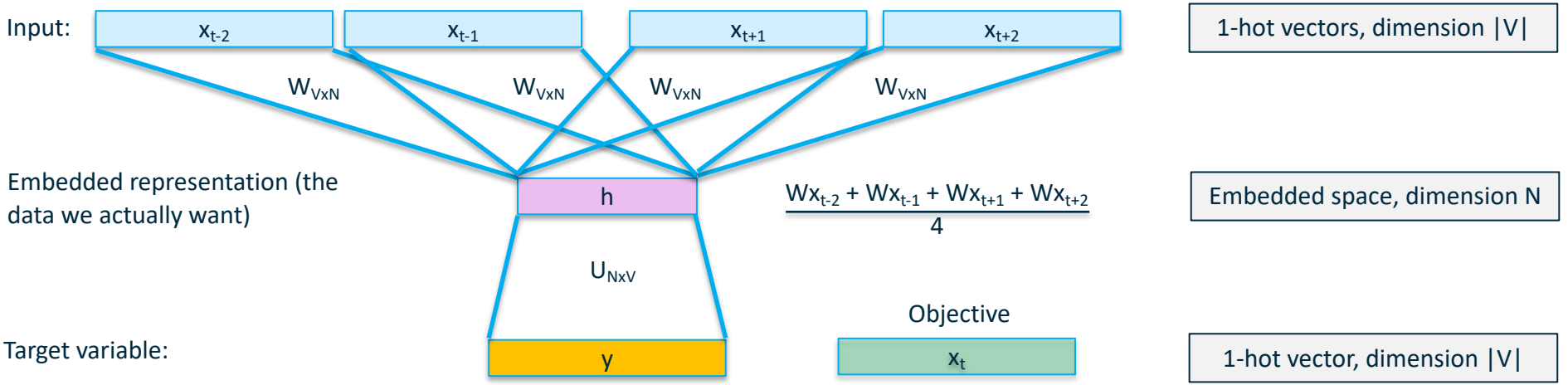
- Principal Component Analysis (PCA)
- Singular Value Decomposition (SVD)
- …

Empirically, denser vectors work better than sparse vectors for many tasks

- E.g. vectors of 300 dimensions, instead of |V| dimensions

Better vector representations: word2vec
1. Continuous Bag Of Words (CBOW)

Choose a context size, c (e.g. c=2), desired dimensionality N (typically N<< |V|)

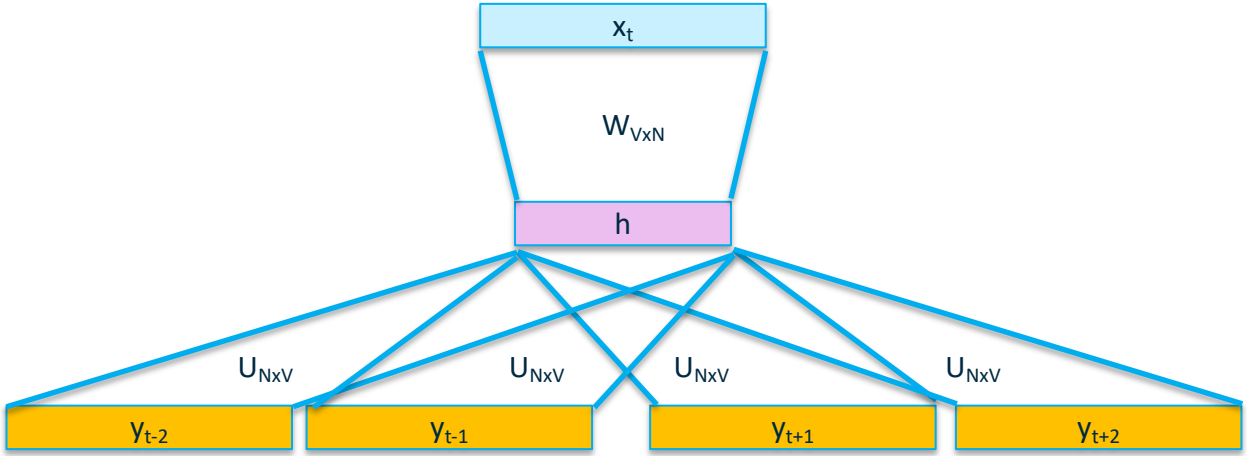


x_0 x_0 x_0 x_1 x_2 x_3 x_4

["Kick", "the", "ball", "into", "the", "opponent", "'s", "goal", "."]

Better vector representations: word2vec
2. Skip-gram

Instead of predicting a word from context, predict context from a word



1-hot vector, dimension |V|

Embedded space, dimension N

1-hot vectors, dimension |V|

x_0 x_1 x_2 x_3 x_4

["Kick", "the", "ball", "into", "the", "opponent", "'s", "goal", "."]

Better vector representations: word2vec

Represented how? This is a non-trivial, empirical question!

After training a model:

- (Smaller) embedded space contains enough information to do well on task
- Averaging of words forces “related” words to have similar embeddings in h
  - Denser word vectors with “more desirable properties” than 1-hot
    - Agree with *some* intuitions about “good” vector representations
    - Better results shown in wide range of NLP tasks
    - Can train *once* on vast amounts of unstructured text

Word2vec word similarities

Can compare vectors by cosine similarity [couldn't in 1-hot: always zero!]

- E.g. which vectors have highest cosine similarity to "japan":

Syntactic	Semantic
Syntactic	Semantic
Syntactic	Semantic
Syntactic	Semantic
Syntactic	Semantic
Syntactic	Semantic
Syntactic	Semantic
Syntactic	Semantic
Syntactic	Semantic
Syntactic	Semantic

```
[('japanese', 0.6607722043991089),
 ('tokyo', 0.6265655755996704),
 ('america', 0.6033484935760498),
 ('europe', 0.5962790250778198),
 ('germany', 0.5782293081283569),
 ('chinese', 0.5763071775436401),
 ('india', 0.5745143294334412),
 ('hawaii', 0.5731386542320251),
 ('usa', 0.5680993795394897),
 ('korea', 0.5648359060287476)]
```

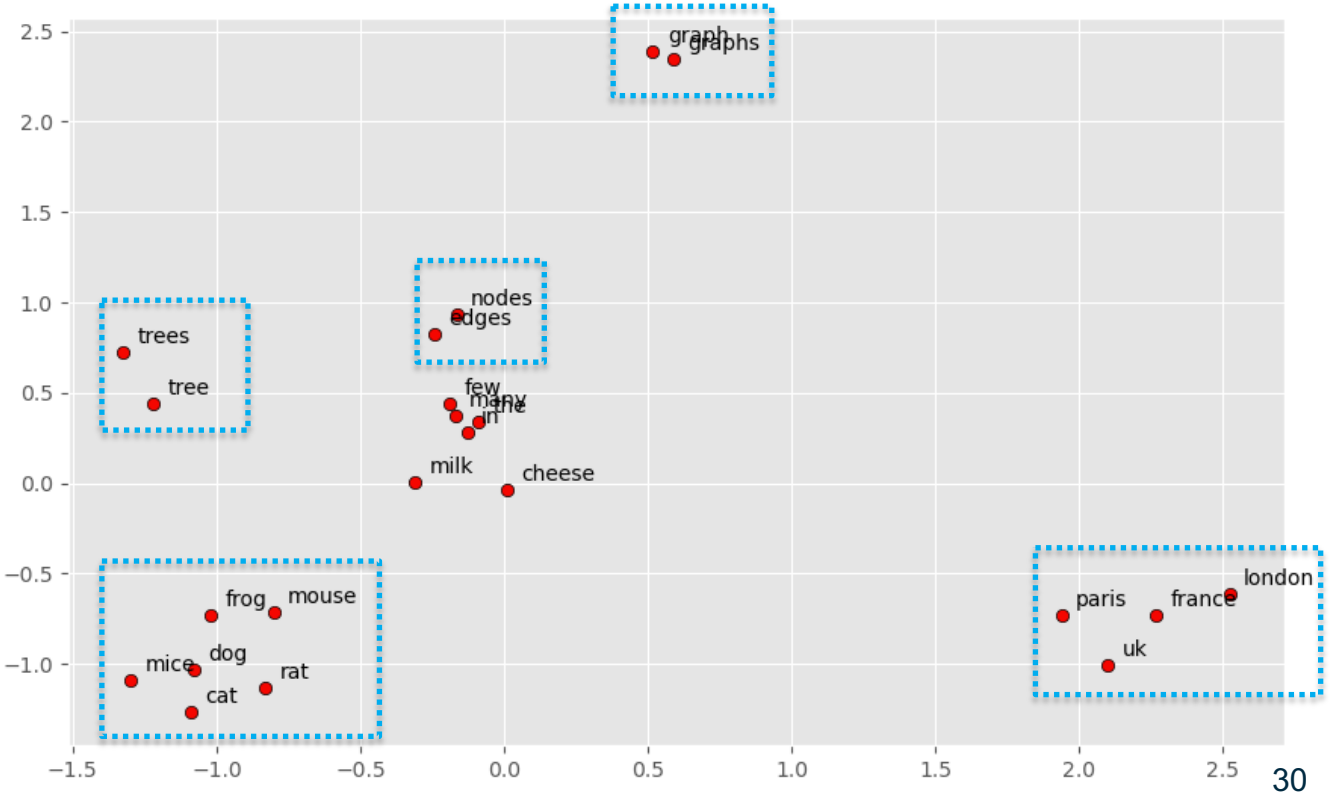
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Data: GoogleNews-vectors-negative300.bin

Vector similarity

Projection into 2D

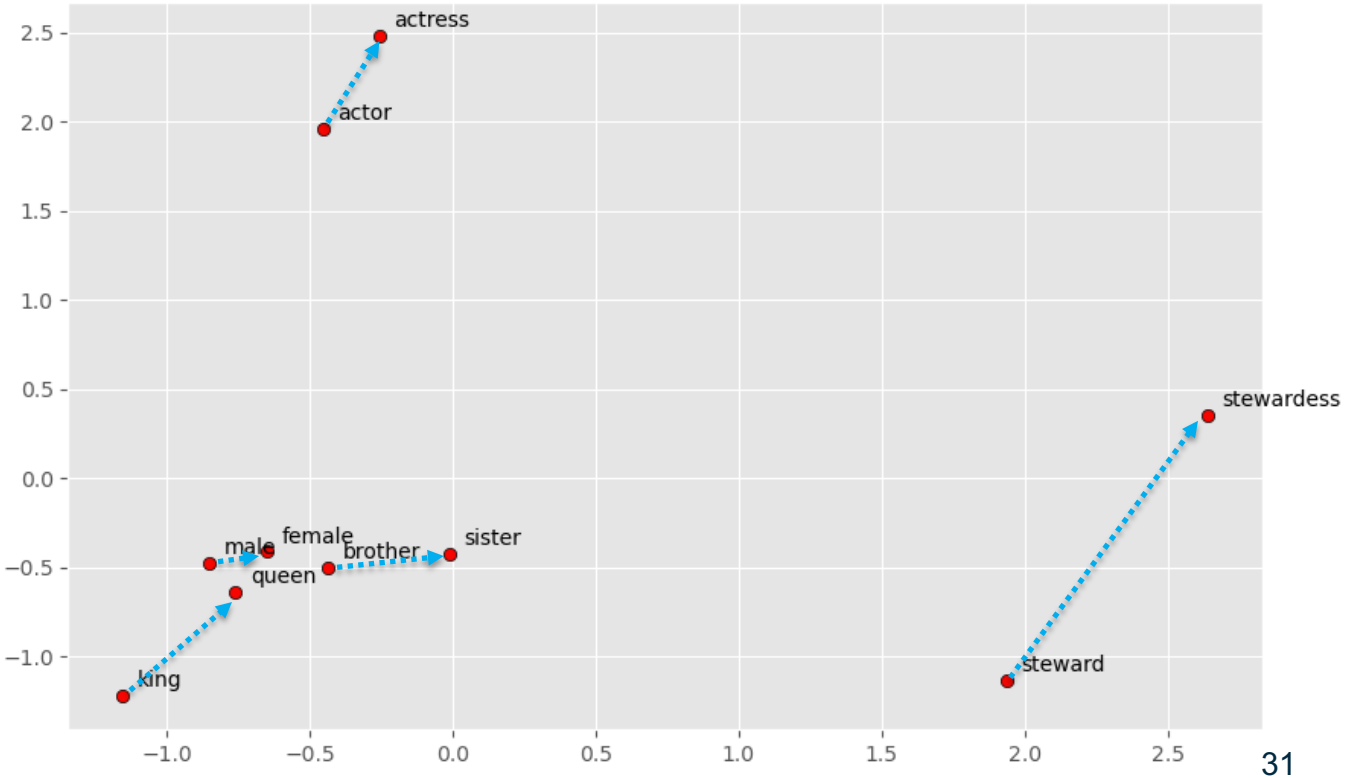
(Example using Principal Component Analysis)



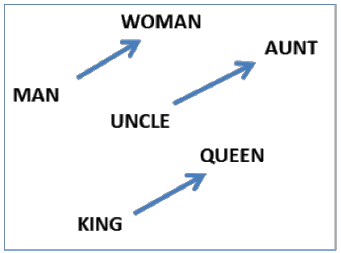
Vector similarity

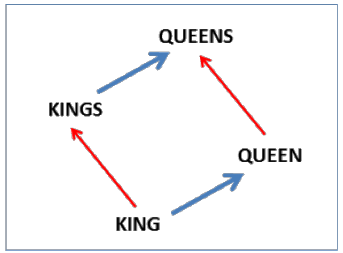
Projection into 2D

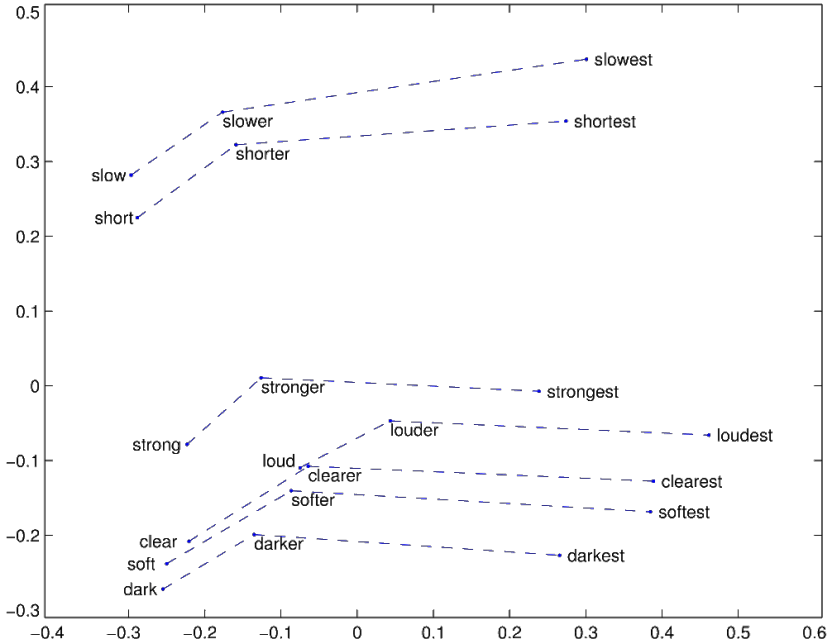
(Example using Principal Component Analysis)



Embedding properties







- Embeddings can capture relational meaning
- e.g., $e_{king} - e_{man} + e_{woman} \approx e_{queen}$
- e.g., $e_{Paris} - e_{France} + e_{Italy} \approx e_{Rome}$

Durham University

Word2vec analogies

Observation/claim: $e_1 - e_2 \approx e_3 - e_4$

Vector algebra (sometimes) fits with analogical reasoning

E.g. brother ~ son

sister ~ _____

$e_{son} - e_{brother} + e_{sister}$

$= e_y$

$\max_{x \in V}(\text{similarity}(e_y, e_x))$

=> x = daughter

	Type of relationship	e_1	e_2	e_3	e_4
		Word Pair 1		Word Pair 2	
Semantic	Common capital city	Athens	Greece	Oslo	Norway
	All capital cities	Astana	Kazakhstan	Harare	Zimbabwe
	Currency	Angola	kwanza	Iran	rial
	City-in-state	Chicago	Illinois	Stockton	California
	Man-Woman	brother	sister	grandson	granddaughter
Syntactic	Adjective to adverb	apparent	apparently	rapid	rapidly
	Opposite	possibly	impossibly	ethical	unethical
	Comparative	great	greater	tough	tougher
	Superlative	easy	easiest	lucky	luckiest
	Present Participle	think	thinking	read	reading
	Nationality adjective	Switzerland	Swiss	Cambodia	Cambodian
	Past tense	walking	walked	swimming	swam
	Plural nouns	mouse	mice	dollar	dollars
	Plural verbs	work	works	speak	speaks

Mikolov et al, Efficient Estimation of Word Representations in Vector Space

Embedding biases

- Embeddings reflect cultural biases
- $e_{man} - e_{woman} \approx e_{computer\ programmer} - e_{homemaker}$
- Male names associated with math while female names with the arts
- Words for specific ethnicities associated with various stereotypes
- Embeddings for adjectives pertaining to “competence” are biased in favor of men but this bias is slowly decreasing
- Embeddings can be used as a historical tool to study bias or change

- Man is to computer programmer as woman is to homemaker? debiasing word embeddings
- Word embeddings quantify 100 years of gender and ethnic stereotypes
- Semantics derived automatically from language corpora contain human-like biases
- Domain-Specific Sentiment Lexicons for Computational Social Science

Even better vector representations

Contextualized word vectors:

Similar words: “plants”, “flora”, …

Similar words: “graphs”, “nodes”, …

- “lots of tall trees in this forest” vs “trees are connected acyclic graphs”





  - Some semantic overlap, but different meanings, connotations, contexts
  - Word2vec always represents the same token using an identical vector
- ELMO, BERT, GPT-2, …
  - Vector representation of a token *also* depends on its context